

Chapter 1

Solutions to selected exercises

Chapter 1

1. Complete the following.

(a) Write in tabular form:

i. $A = \{x: x \text{ is a letter in "do not repeat"}\}$

$$A = \{d, o, n, t, r, e, p, a\}$$

ii. $3\mathbb{Z} = \{x \in \mathbb{Z}: x = 3n \text{ for some } n \in \mathbb{Z}\}$

$$3\mathbb{Z} = \{\dots, -6, -3, 0, 3, 6, \dots\}$$

iii. $N = \{x: x \text{ is odd and } x \text{ is even}\}$ (recall that 0 is even)

$$N = \emptyset$$

(b) Write these sets in set-builder form:

i. Let $A = \{3, 4, 5, 6, 7, \dots\}$

$$A = \{x \in \mathbb{Z}: x \geq 3\}$$

ii. Let $B = \{3, 5, 7, 9, \dots\}$

$$B = \{x: x = 2n + 1 \text{ where } n \in \mathbb{Z} \text{ and } n \geq 1\}$$

iii. Let $\mathbb{Q}$ be the set of rational numbers.

$$\mathbb{Q} = \{x: x = \frac{a}{b} \text{ where } a, b \in \mathbb{Z}\}$$

2. Find all subsets of $A = \{1, 2, 3\}$.

$\{1, 2, 3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1\}, \{2\}, \{3\}, \emptyset$. There are 3 elements in A so there are $2^3 = 8$ subsets.

3. Prove that every nonempty subset has a proper subset.

PROOF Let A be any non-empty subset. $\emptyset \subseteq A$ since $\emptyset$ is a subset of every set, and $\emptyset \subset A$ (a proper subset) since $A \neq \emptyset$.

4. Prove that if A is a subset of $\emptyset$, then $A = \emptyset$.

PROOF Let $A \subseteq \emptyset$. $\emptyset \subseteq A$ for any subset A . So then $A = \emptyset$.

5. Prove that $A = \{2, 3, 4, 5\}$ is not a subset of $B = \{x \mid x \text{ is even}\}$.

PROOF 5 is not even since $5 = 2 \cdot 2 + 1$. So then $5 \notin B$. Thus $A \not\subseteq B$.

6. Prove that A and B are subsets of $A \cup B$.

PROOF Let $a \in A$. $a \in A$ so then $a \in A \cup B$ by definition of the union of sets. Thus $A \subseteq A \cup B$. Similarly we can show $B \subseteq A \cup B$.

7. Prove that $A = A \cup A$ and $A = A \cap A$.

PROOF ($A = A \cup A$). By exercise 6, $A \subseteq A \cup A$. So we need only show that $A \cup A \subseteq A$. Let $a \in A \cup A$. Then $a \in A$ or $a \in A$, implying that $A \cup A \subseteq A$. Thus $A \cup A = A$.

($A = A \cap A$). Let $a \in A$. Then $a \in A$ and $a \in A$. Thus $A \subseteq A \cap A$. Let $a \in A \cap A$. Then $a \in A$ and $a \in A$ by definition, and this implies that $A \cap A \subseteq A$.

Thus $A = A \cap A$.

8. Prove that $A \cap B$ is a subset of A and of B .

PROOF Let $x \in A \cap B$. Then $x \in A$ and $x \in B$ by definition. So then $x \in A$ implying that $A \subseteq A \cap B$. Similarly we can show that $B \subseteq A \cap B$.

9. Prove that $A \cup B = \emptyset$ implies that $A = \emptyset$ and $B = \emptyset$.

PROOF Suppose that $A \cup B = \emptyset$. $A \subseteq A \cup B = \emptyset$ by exercise 6. Thus $A \subseteq \emptyset$, and this implies that $A = \emptyset$ by exercise 4.

Similarly we can show that $B = \emptyset$.

10. Prove that $A \cup B = B$ if and only if $A \subseteq B$.

PROOF ($\Rightarrow$) Let $A \cup B = B$. $A \subseteq A \cup B = B$ (by exercise 6) which implies that $A \subseteq B$.

($\Leftarrow$) Let $A \subseteq B$. $B \subseteq A \cup B$ (again exercise 6) so we only need to show that $A \cup B \subseteq B$. Let $a \in A \cup B$ then $a \in A$ or $a \in B$. If $a \in B$ then $A \cup B \subseteq B$ as required. If $a \in A$ then $A \subseteq B$ implies that $a \in B$. Thus $A \cup B \subseteq B$. Which implies that $A \cup B = B$.

- For “if and only if/ $\iff$ ” statements, you need to show both directions, $\Rightarrow$ and $\Leftarrow$.
- For $A = B$ statements, it is usually easiest to show that $A \subseteq B$ and $B \subseteq A$ separately.

11. Prove that $A \cap B = B$ if and only if $B \subseteq A$.

PROOF ($\Rightarrow$) Let $A \cap B = B$. $B = A \cap B \subseteq A$ by exercise 8 so then $B \subseteq A$.

($\Leftarrow$) Let $B \subseteq A$. We already know that $A \cap B \subseteq B$ by ex. 8. So we need only show that $B \subseteq A \cap B$. Let $b \in B$. By assumption $B \subseteq A$ so then $b \in A$ which implies that b is in A and B , i.e., $b \in A \cap B$.

12. Prove that $A \cup (A \cap B) = A$.

PROOF $A \cup (A \cap B) = A$ if and only if $A \cap B \subseteq A$ by ex. 10, and $A \cap B \subseteq A$ by ex. 8.

13. Prove that $A \cap (A \cup B) = A$.

PROOF $A \cap (A \cup B) = A$ if and only if $A \subseteq A \cup B$ by ex. 11, and $A \subseteq A \cup B$ by ex. 6.

14. Prove that $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$.

PROOF Let $P \in \mathcal{P}(A \cap B)$. Then $P \subseteq A \cap B$ implying that $P \subseteq A$ and $P \subseteq B$ thus $P \in \mathcal{P}(A) \cap \mathcal{P}(B)$.

Now let $P \in \mathcal{P}(A) \cap \mathcal{P}(B)$. Thus $P \subseteq A$ and $P \subseteq B$ which means that $P \subseteq A \cap B$ and that $P \in \mathcal{P}(A \cap B)$.

Chapter 2

Prove or disprove the following.

1. A constant function is not one-to-one if the domain has more than one element.

TRUE. Let $f: A \rightarrow B$ be a constant function and $|A| > 1$. Since $|A| > 1$ it has at least two elements, say a_1 and a_2 . f is constant so then $f(a_1) = f(a_2) = b$ for some $b \in B$. Thus f is not one-to-one.

Note that if the domain has only one element then this statement is false. Let $A = \{a\}$ and $f: A \rightarrow B$ and $f(a) = b$ for all $a \in A$. If $f(a_1) = f(a_2)$ then $a_1 = a_2$ because A only has one element. So f is constant and also one-to-one. In general constant functions are not 1-1, i.e., the statement “constant functions are 1-1” is false.

- (a) The identity function is one-to-one.

TRUE. Let $f : A \rightarrow A$ be the identity function. If $f(a_1) = f(a_2)$ then $f(a_1) = a_1 = a_2 = f(a_2)$ because $f(a) = a$ for any $a \in A$. Thus f is one-to-one.

(a) The identity function is onto.

TRUE. Let $f : A \rightarrow A$ be the identity function and $a \in A$. $f(a) = a$ so then f is onto.

Let $g : A \rightarrow B$ and $f : B \rightarrow C$. Prove that if $f \circ g : A \rightarrow C$ is one-to-one and g is onto then f is also one-to-one.

PROOF Let $b_1, b_2 \in B$ such that $b_1 \neq b_2$. Because g is onto there is a $a_1, a_2 \in A$ such that $g(a_1) = b_1$ and $g(a_2) = b_2$. So then $f(b_1) \neq f(b_2)$ since $f(g(a_1)) \neq f(g(a_2))$.

Suppose that $f : A \rightarrow B$ is a one-to-one function. Prove or disprove that if $f^{-1} : f(B) \rightarrow A$ is a function then f^{-1} is also one-to-one.

TRUE. Recall: It is shown in the notes that for a function g , the inverse of g is a function if and only if g is 1-1. Since f^{-1} is a function and $(f^{-1})^{-1} = f$ is also a function, it follows that f^{-1} is 1-1.

Alternatively,

PROOF Suppose to the contrary that f^{-1} is *not* 1-1. So then there is $b_1, b_2 \in B$ such that $b_1 \neq b_2$ and $f^{-1}(b_1) = f^{-1}(b_2) = a$ for some $a \in A$. Thus $f(f^{-1}(b_1)) = b_1 \neq b_2 = f(f^{-1}(b_2))$ which contradicts the assumption that f is a function.

Let $f : A \rightarrow B$ be a one-to-one function. Show that $|A| \leq |B|$.

PROOF Let $|A|$ be the number of items and $|B|$ be the number of boxes. By the pigeon hole theorem, if $|A| > |B|$ then f would map (or place) at least 2 items to the same box and thus not be 1-1. Thus f must be 1-1.

Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions. Prove that $g \circ f : A \rightarrow C$ is a function.

PROOF f is a function implies that for any $a \in A$ there is a unique $b \in B$ such that $f(a) = b$. Also, g is a function implies that for any $b \in B$ there is a unique $c \in C$ such that $g(b) = c$. Thus for any $a \in A$ there is a unique $c \in C$ such that $g(f(a)) = g(b) = c$. So $g \circ f$ is a function.

(This problem was dropped) Let $f : A \rightarrow A$, $g : A \rightarrow A$, and $h : A \rightarrow A$; prove that $(h \circ g) \circ f = h \circ (g \circ f)$.

PROOF Let $a_0 \in A$. Since f, g, h are functions there are $a_1, a_2, a_3 \in A$ such that, $f(a_0) = a_1$, $g(a_1) = a_2$, and $h(a_2) = a_3$. $((h \circ g) \circ f)(a_0) = h(g(f(a_0))) = h(g(a_1)) = h(a_2) = a_3$. Also, $(h \circ (g \circ f))(a_0) = h((g \circ f)(a_0)) = h(g(f(a_0))) = h(g(a_1)) = h(a_2) = a_3$. Thus $(h \circ g) \circ f = h \circ (g \circ f)$.

Prove that if $f : A \rightarrow B$ is onto and $g : B \rightarrow C$ is onto, then $(g \circ f) : A \rightarrow C$ is onto.

PROOF Let $c \in C$. g is onto so then there exists a $b \in B$ such that $g(b) = c$. Since f is onto there also exists an $a \in A$ such that $f(a) = b$. So then $g(f(a)) = g(b) = c$; thus $g \circ f$ is onto.

Each of the following statements, $P(x, y)$, define a relation R in the natural numbers, $\mathbb{N}$, i.e., $R = (\mathbb{N}, \mathbb{N}, P(x, y))$. Show whether each of the following relations is reflexive.

1. $P(x, y) = "x \text{ is less than or equal to } y"$

REFLEXIVE: Let $x \in \mathbb{N}$. $x = x \Rightarrow x \leq x$ so then R is reflexive.

(a) $P(x, y) = "x \text{ divides } y"$

REFLEXIVE: Let $x \in \mathbb{N}$. $x = 1 \cdot x$ so then $x|x$. Thus R is reflexive.

(a) $P(x, y) = "x + y = 10"$

NOT REFLEXIVE: Consider the natural number 4; $4 + 4 \neq 10$. Thus R is not reflexive.

(a) $P(x, y) = "x \text{ and } y \text{ are relatively prime}"$

NOT REFLEXIVE: Consider the natural number 8; $\gcd(8, 8) = 8 \neq 1$ since $8|8$. Thus R is not reflexive.